

ML Assignment

Shashank Peddi
BT2024210

1 Introduction

The objective of this assignment is to build polynomial regression models for the given geothermal energy datasets. The input variables describe different operational or spatial parameters, while the target variable represents the corresponding output score.

Two problems were considered:

- **Phase 1:** Power Plant Steam Turbine Optimization
- **Phase 2:** Subterranean Thermal Reservoir Mapping

For both problems, polynomial regression was used. The polynomial degree was selected using cross-validation. The main evaluation metrics used were Mean Squared Error (MSE) and the R^2 score.

2 General Approach

The implementation follows the same basic procedure for both problems.

2.1 Loading the Data

The training data is loaded using **pandas**. The feature columns are separated from the target column y .

For Phase 1, the six features are

$$x_1, x_2, x_3, x_4, x_5, x_6.$$

For Phase 2, there are three features:

$$x_1, x_2, x_3.$$

The test data is kept separate and is used only after selecting the final polynomial degree.

2.2 Generating Polynomial Features

The **PolynomialFeatures** class from scikit-learn is used to generate the polynomial terms. For example, for two variables, a degree-2 model contains terms such as

$$1, \quad x_1, \quad x_2, \quad x_1^2, \quad x_1x_2, \quad x_2^2.$$

Thus, interaction terms are automatically included, as long as their total degree does not exceed the selected degree.

2.3 Finding the Regression Coefficients

The regression coefficients are calculated using the normal equation:

$$\theta = (X^T X)^{-1} X^T y.$$

In the implementation, the pseudoinverse is used instead of directly calculating the inverse:

```
theta = np.linalg.pinv(X_poly) @ y
```

2.4 Polynomial Degree Selection

The polynomial degree was tested over the complete range specified for each problem.

For Phase 1, degrees 1 through 10 were tested.

For Phase 2, degrees 1 through 20 were tested.

Five-fold cross-validation was used for each degree.

The average validation MSE and R^2 were calculated for every degree.

The degree with the lowest cross-validation MSE was selected.

3 Phase 1: Steam Turbine Optimization

3.1 Problem Setup

Phase 1 contains six input parameters. The target variable y represents the Net Power Score.

The model was tested for polynomial degrees from 1 to 10.

3.2 Degree Selection

The cross-validation results are shown in Figure 1.

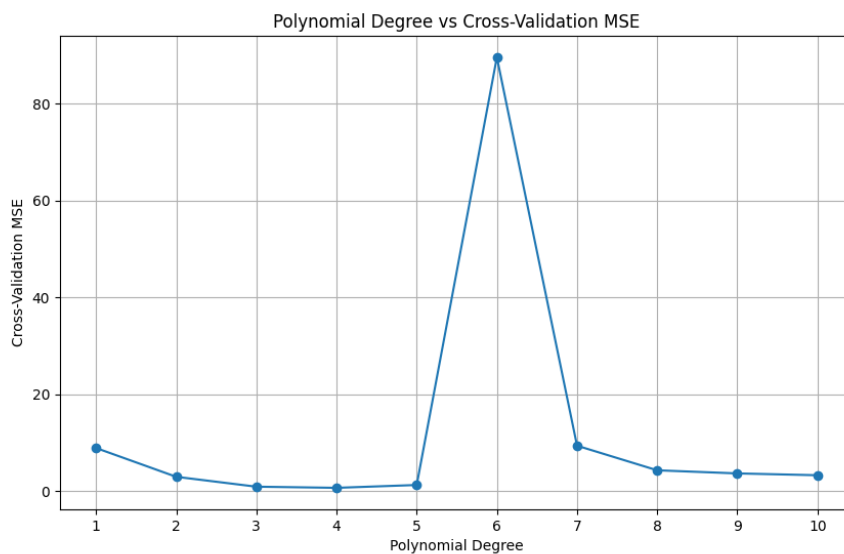


Figure 1: Polynomial degree versus cross-validation MSE.

The lowest cross-validation MSE was obtained with degree 4:

$$\boxed{\text{Degree} = 4}$$

The corresponding validation results were

$$\text{MSE} = 0.727863$$

and

$$R^2 = 0.924730.$$

Therefore, degree 4 was selected for the final Phase 1 model.

3.3 Training Performance

After selecting degree 4, the model was trained again using the complete training dataset. The resulting training performance was:

$$\text{Training MSE} = 0.385723$$

$$R^2 = 0.960850.$$

4 Phase 2: Subterranean Thermal Reservoir Mapping

4.1 Problem Setup

Phase 2 uses three input parameters:

$$x_1, \quad x_2, \quad x_3$$

The target y is the Thermal Anomaly Score.

Since the problem allows polynomial degrees up to 20, all degrees from 1 through 20 were evaluated using five-fold cross-validation.

4.2 Degree Selection

The cross-validation results are shown in Figure 2.

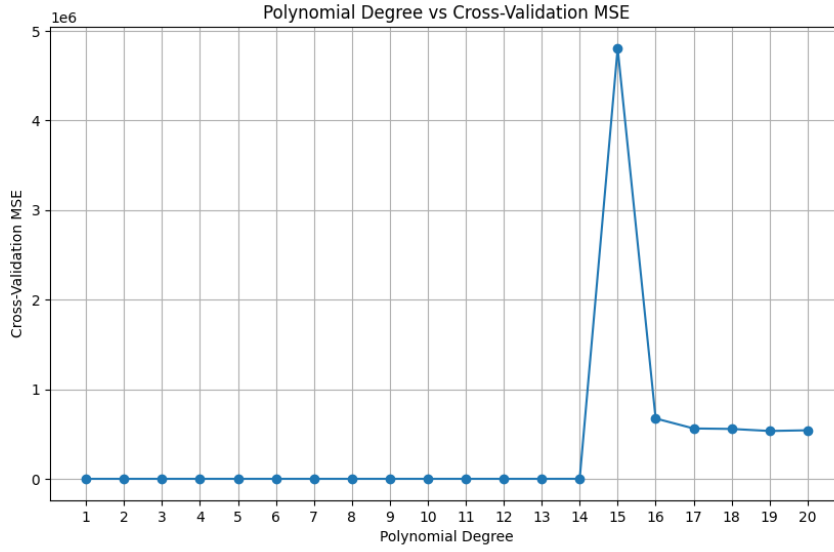


Figure 2: Polynomial degree versus cross-validation MSE.

The complete degree range was tested, but degree 8 produced the lowest cross-validation MSE. Therefore,

$$\boxed{\text{Degree} = 8}$$

was selected.

The validation results for this model were

$$\text{MSE} = 0.286778$$

and

$$R^2 = 0.993448.$$

Degree 9 produced an almost identical result, but its MSE was slightly higher:

$$0.286839 > 0.286778.$$

Therefore, degree 8 was selected based on the minimum validation MSE.

4.3 Training Performance

The final degree-8 model was trained using the complete training set.

The resulting training metrics were:

$$\text{Training MSE} = 0.179923$$

$$R^2 = 0.996068.$$

Models with degrees higher than 8 showed increasingly poor validation performance. In particular, degrees 15–20 produced very large validation MSE values and strongly negative R^2 values. This indicates overfitting, so those models were avoided.

5 Final Models

Based on the analysis, the selected models are summarized in Table 1.

Problem	Degree	Terms	CV MSE	CV R^2
Phase 1	4	210	0.727863	0.924730
Phase 2	8	165	0.286778	0.993448

Table 1: Summary of selected polynomial models

For both problems, the final models were retrained using the complete training dataset after the degree was selected. The final models were then used to generate predictions for the corresponding test datasets.

6 Conclusion

Polynomial regression was used for both problems. The polynomial degree was selected using five-fold cross-validation rather than simply using the maximum allowed degree.

For Phase 1, degree 4 was selected because it achieved the lowest validation MSE of 0.727863.

For Phase 2, degree 8 was selected because it achieved the lowest validation MSE of 0.286778.

The results demonstrate that increasing the polynomial degree does not always improve prediction performance. At higher degrees, the number of polynomial terms increases significantly, and the validation performance can deteriorate. Therefore, cross-validation was used to select a suitable degree for each dataset.